Rev	Date	Eng	Changes
${PROJECTREVISION}	12/15/2023	SD + ES	Adding silkscreen labels on BM ports ("1" and

${PCBPARTNAME}

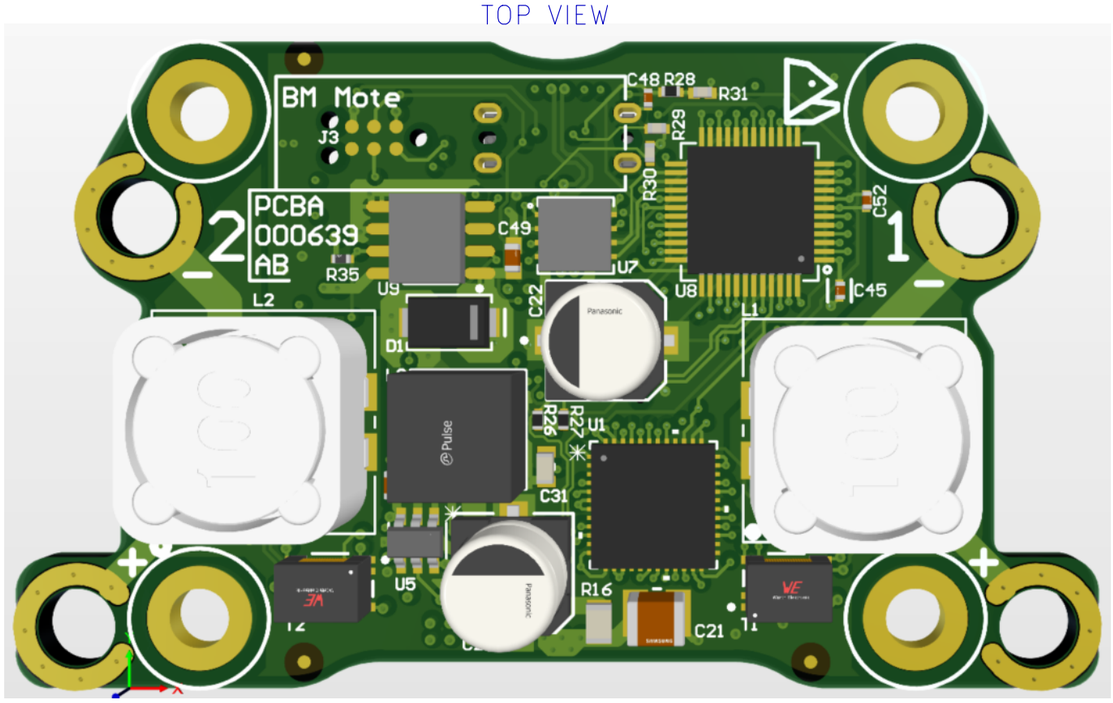
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P/N:${PCBPARTNUMBER}

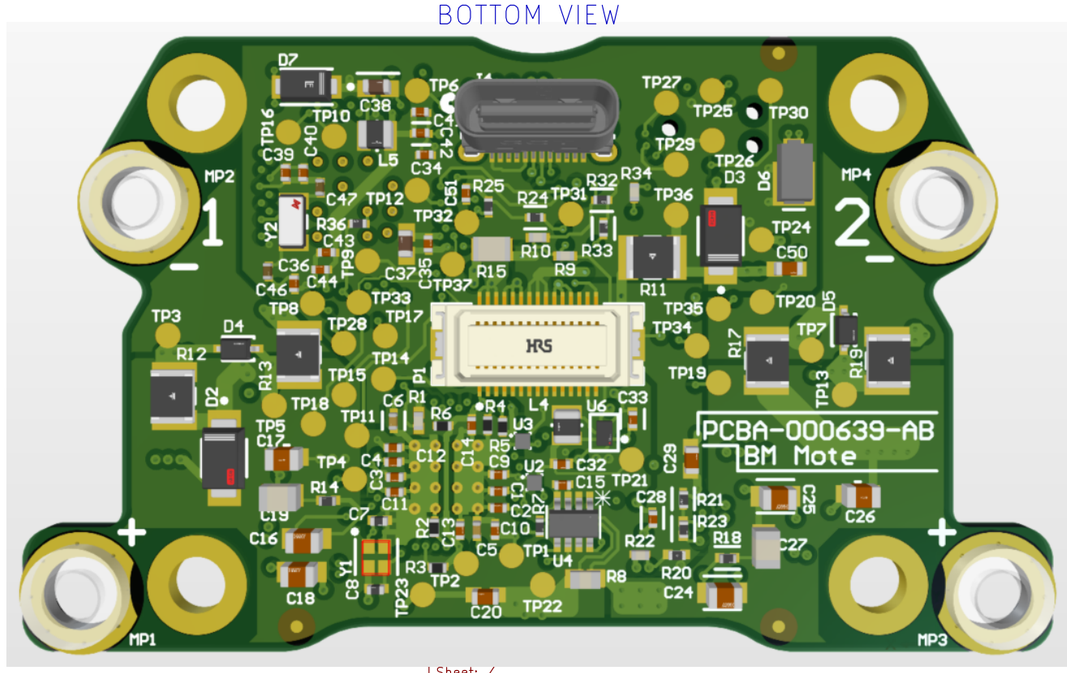
Index

Schematic Document	SheetNumber
BM_Mote_1_Cover.SchDoc	1
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BM_Mote_1_Load.SchDoc	4
BM_Mote_1_PoDL.SchDoc	5
BM_Mote_1_PowerMon.SchDoc	6
BM_Mote_1_Power.SchDoc	7
BM_Mote_1_Processor.SchDoc	8
BM_Mote_1_USB.SchDoc	9

TOP VIEW



BOTTOM VIEW



I2c Address Table

Device	I2C Address
Power Monitor	0x41

DIMENSIONS: 49 mm x 37 mm

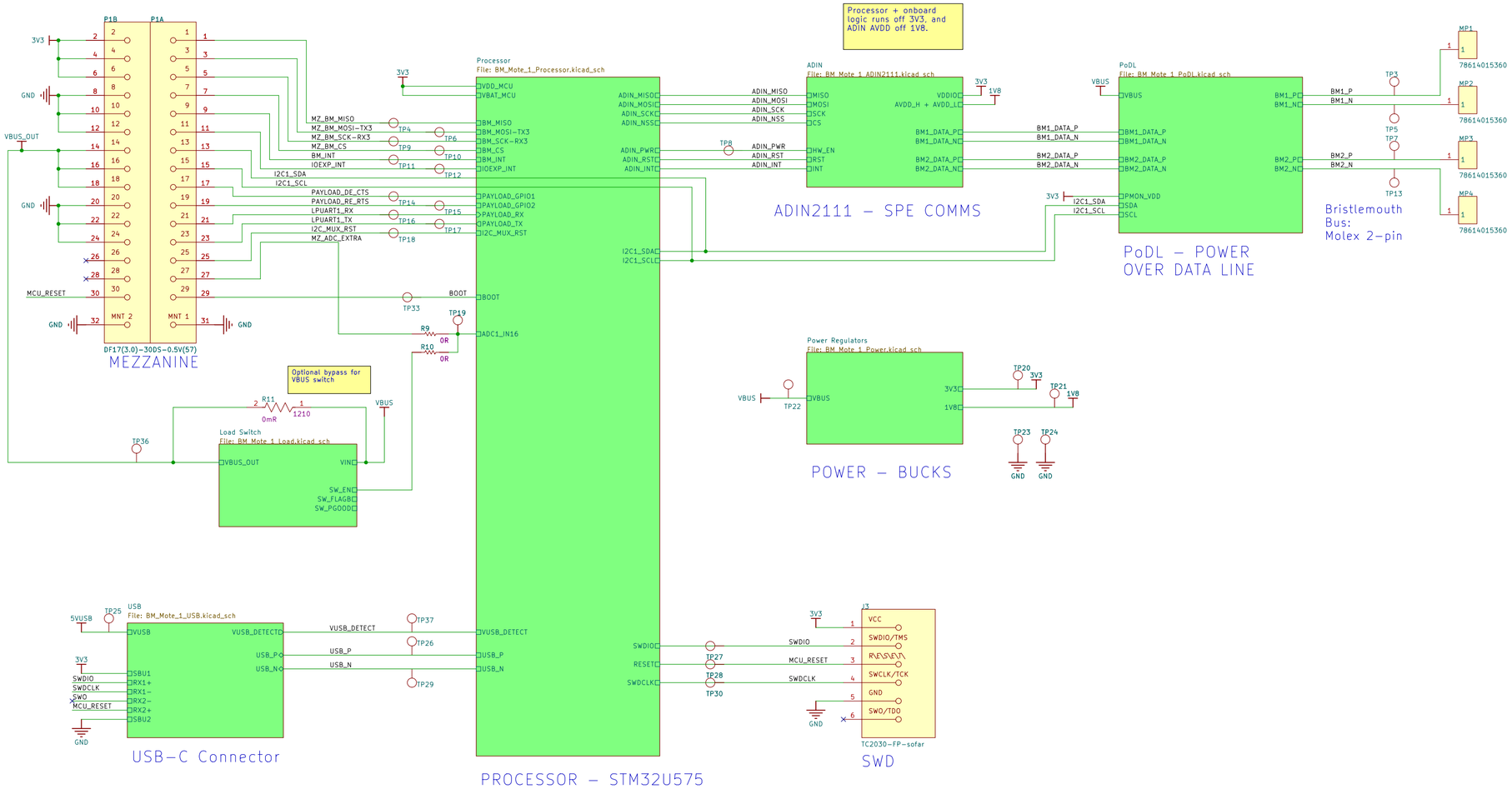
Top-Level Schematic
File: BM_Mote_1_Master.kicad_sch



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https://creativecommons.org/licenses/by-nd/4.0/legalcode

Sheet: /		
File: BM_Mote_000639-AB.kicad_sch		
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Size: A3	Date:	Rev: =ProjectRevision
KiCad E	${PCBPARTNUMBER}	Id: 1/9

=ProjectAuthor



Top-Level

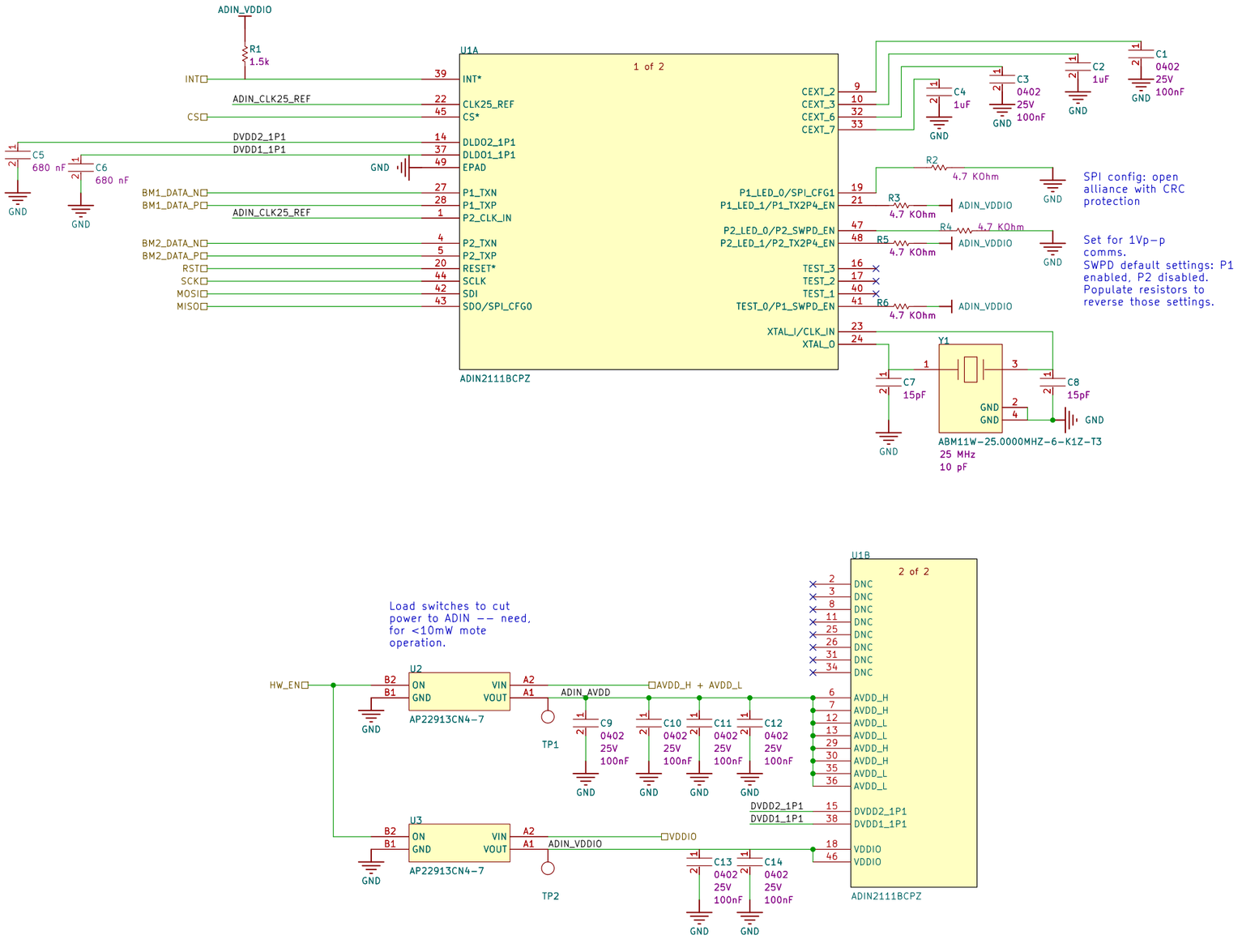
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Title: Top-Level	
Size: A2 Date:	**Rev: =ProjectRevision**
KiCad E.D.A. 10.0.6	Id: 2/9

ADIN2111 Single-Pair Ethernet

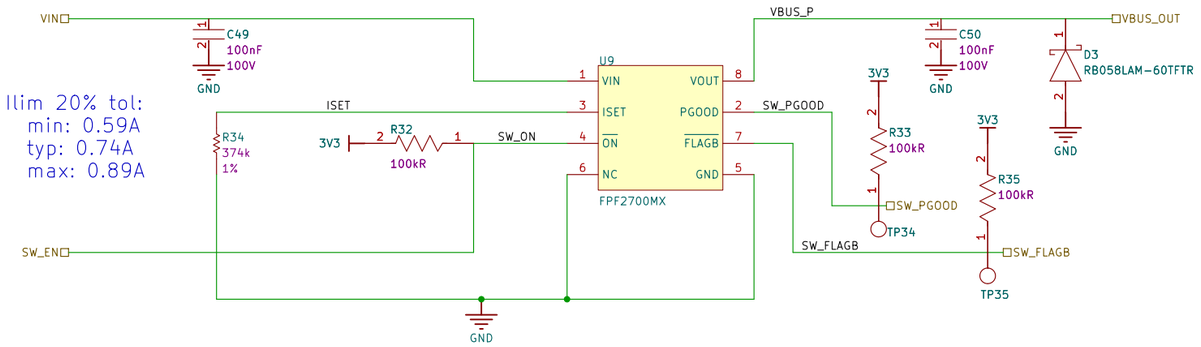


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Size: A3	Date:
KiCad E.D.A. 10.0.6	${PCBPARTNUMBER}
Rev: =ProjectRevision	
Id: 3/9	

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Load Switch for VBUS



Sheet: /Top-Level Schematic/Load Switch/	
File: BM_Mote_1_Load.kicad_sch	
Title: Load Switch	Load Switch
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KiCad E.D.A. 10.0.6	**Rev: =ProjectRevision**
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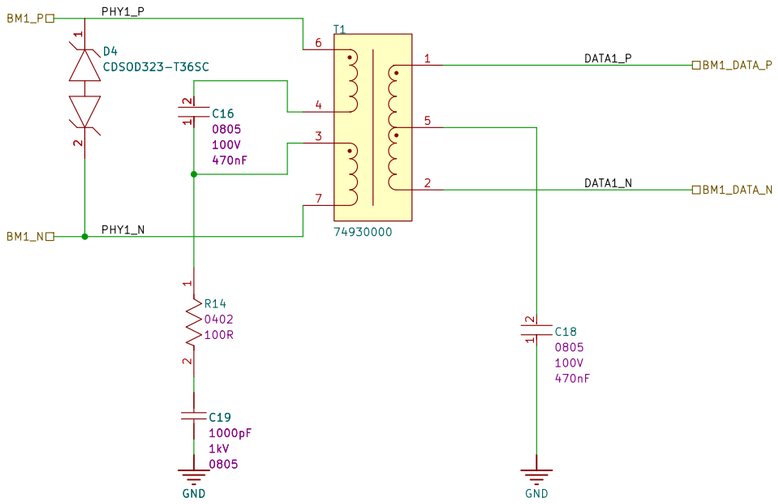
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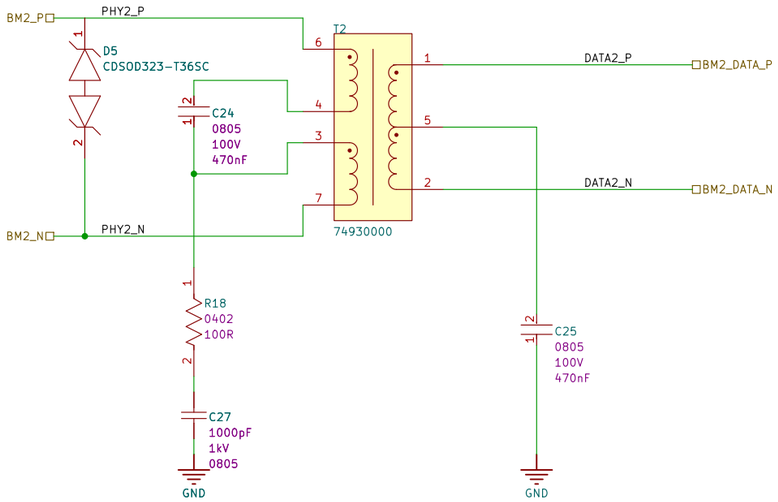
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Power Over Dataline

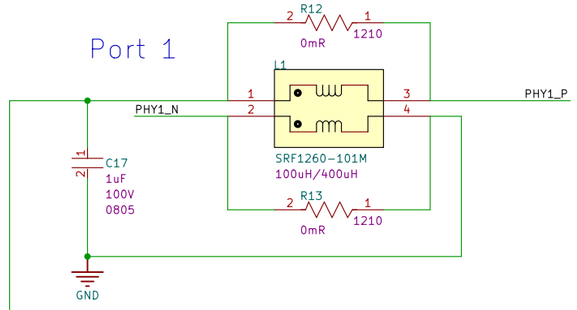
Coupling Network – Port 1



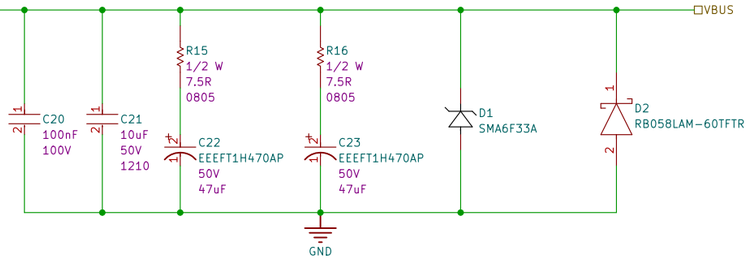
Coupling Network – Port 2



Port 1



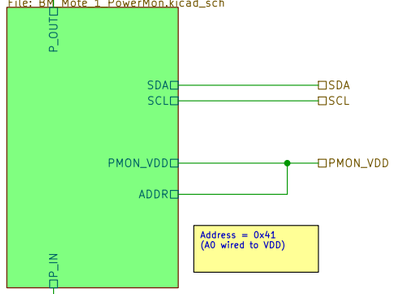
DC Connection + Circuit protections



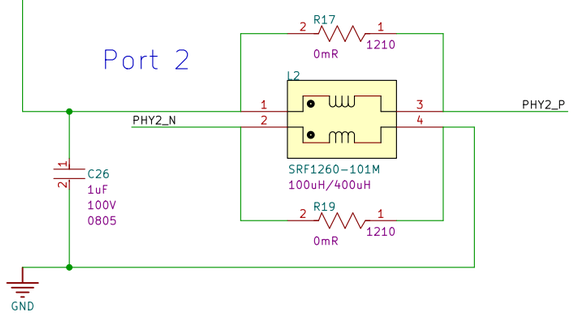
Snubber resistors must be at least 1/2 W, to support hot-plugging.

Power Monitor

File: BM_Mote_1_PowerMon.kicad_sch



Port 2



Sheet: /Top-Level Schematic/PoDL/
File: BM_Mote_1_PoDL.kicad_sch

Title: PoDL (Power over Data Line) PoDL (Power over Data Line)

Size: A3	Date:	**Rev: =ProjectRevision**
KiCad E.D.A. 10.0.6		Id: 5/9

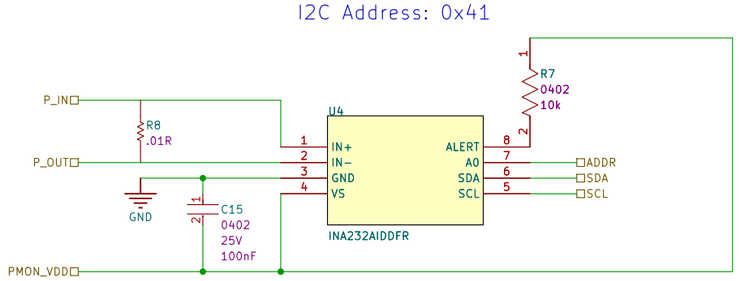
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I2C Address: 0x41



Power Monitor

Sheet: /Top-Level Schematic/PoDL/Power Monitor/
File: BM_Mote_1_PowerMon.kicad_sch

Title: Bus Power Monitor Bus Power Monitor

Size: A3 Date:

Rev: =ProjectRevision

KiCad E.D.A. 10.0.6

${PCBPARTNUMBER}

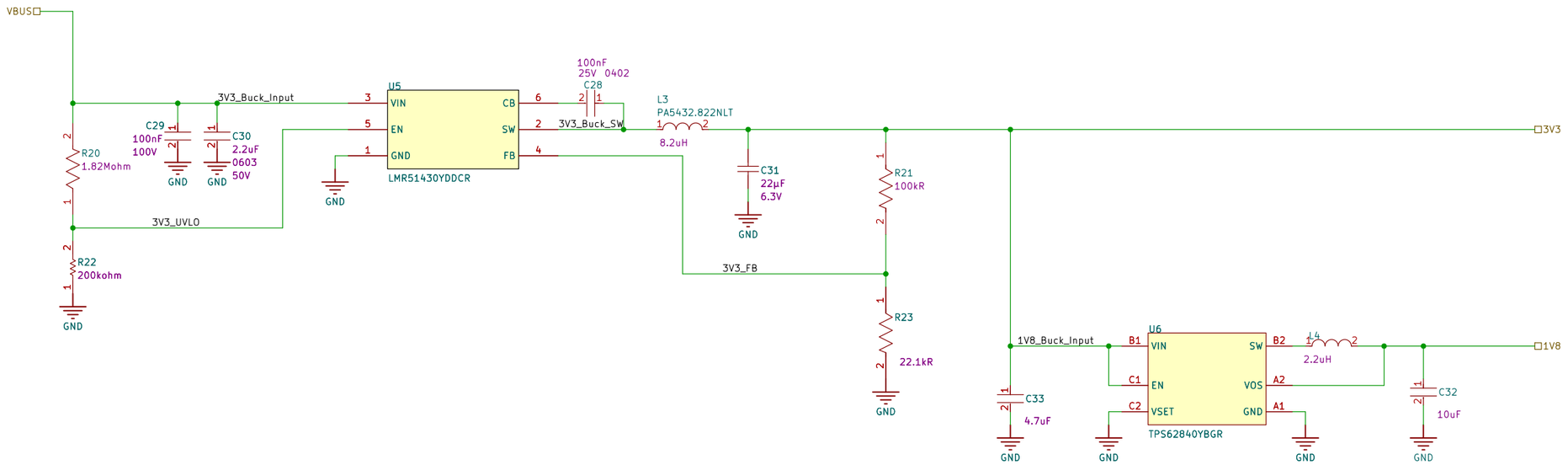
Id: 6/9

=ProjectR

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=ProjectAuthor

Buck Converters – 3V3 and 1V8



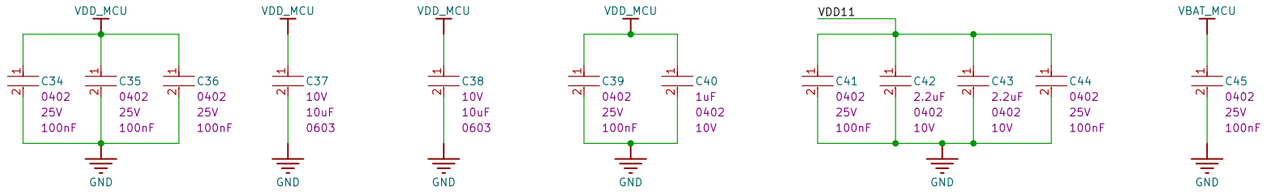
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KiCad E.D.A. 10.0.6	${PCBPARTNUMBER}	Id: 7/9

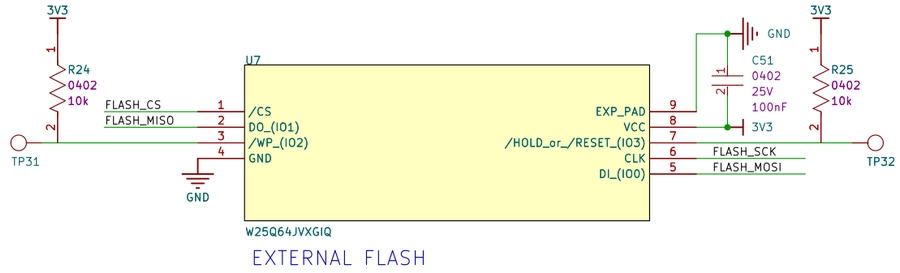
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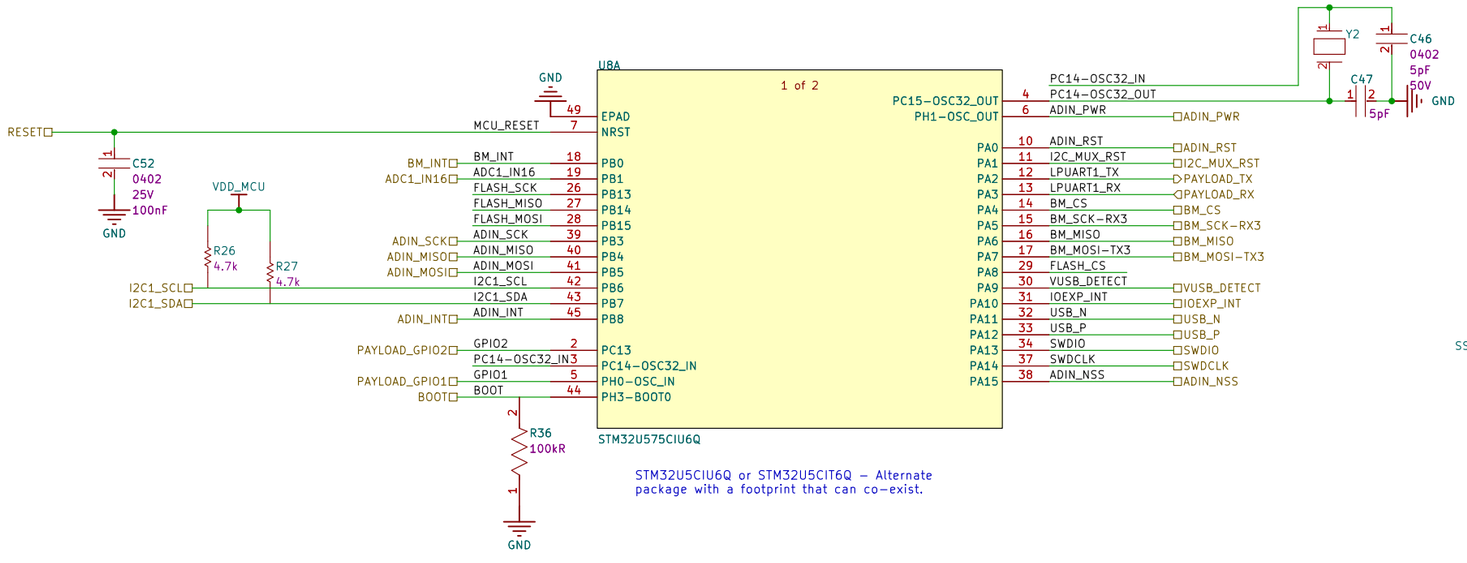
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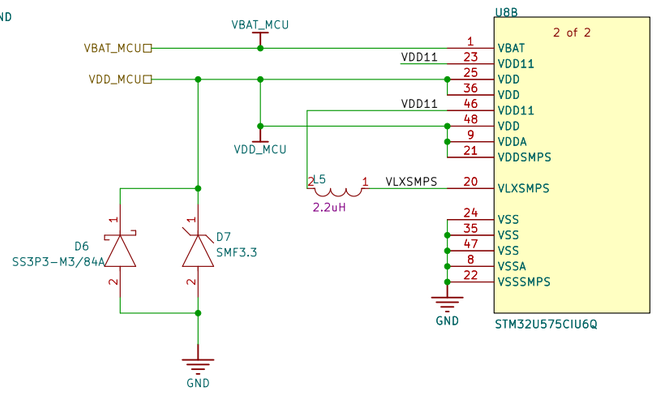
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STM32U575 Processor + External Flash









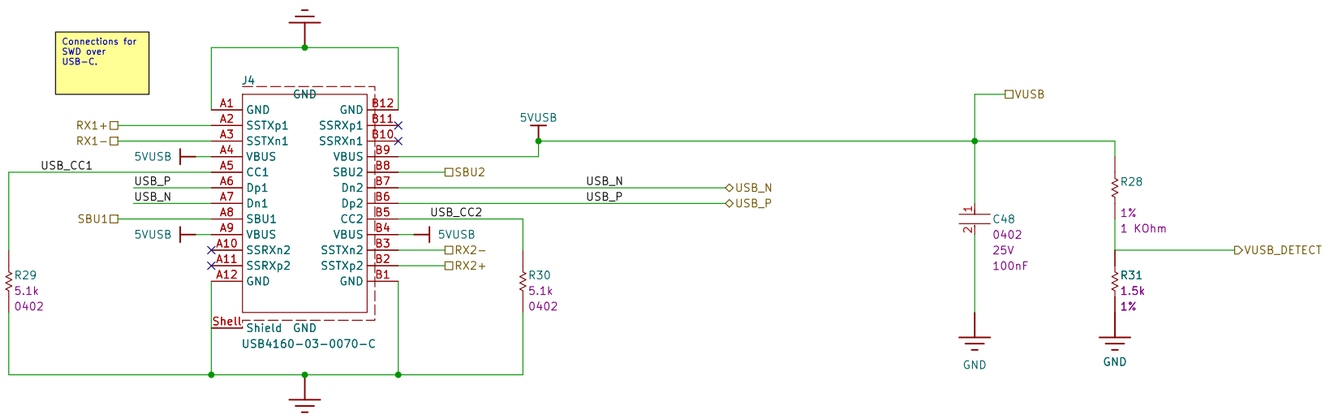
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Size: A3	Date:
KiCad E.D.A. 10.0.6	${PCBPARTNUMBER}
Rev: =ProjectRevision	
Id: 8/9	

=ProjectR

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=ProjectAuthor

USB-C Connector



SWD-over-USB using DEBUGOTRON board: https://github.com/alvarop/debugotron/releases

Sheet: /Top-Level Schematic/USB/	
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KiCad E.D.A. 10.0.6	${PCBPARTNUMBER}
Rev: =ProjectRevision	Id: 9/9

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